Supply Chain Optimization and Strategic Growth Analysis

Technical Documentation

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This documents provides a complete technical walkthrough of the data analysis process, including all code, methodologies, and implementation details.

1. Technical Overview

1.1 Tools and Technologies

This analysis was conducted using: - **Python 3.x** - Primary programming language - **Pandas** - Data manipulation and analysis - **Matplotlib & Seaborn** - Data visualization - **NumPy** - Numerical computations

1.2 Data Sources

The analysis utilized seven CSV files: 1. `sales.csv` - Transactional sales data (770,109 records) 2. `products.csv` - Product specifications 3. `purchase_prices.csv` - Procurement cost data 4. `refunds.csv` - Product return records 5. `region_to_state_distance.csv` - Geographic mapping 6. `regions.csv` - Regional identifiers 7. `continental_us_state_centers.csv` - State coordinates for mapping

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2. Environment Setup and Data Loading

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3. Data Processing and Transformation

3.1 Temporal Data Parsing

Converting date strings to datetime objects enables time-series analysis and temporal aggregation.

3.2 Geospatial Mapping

Challenge: Sales data is at the state level, while purchase data is at the region level.

Solution: Map each state to its nearest region using the distance matrix.

3.3 Cost Calculation and Profit Analysis

Methodology: 1. Calculate unit cost from purchase data 2. Map region IDs to region names 3. Merge cost data with sales data 4. Handle missing values using hierarchical imputation

3.4 Data Imputation Strategy

Problem: ~40% of sales transactions lack matching purchase price records.

Solution: Hierarchical imputation using national averages as fallback.

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4. Visualization Generation

4.1 Financial Trends Analysis

4.2 Product Performance Analysis

4.3 Refund Rate Analysis

4.4 Geospatial Visualization

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5. Statistical Summary

5.1 Descriptive Statistics

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6. Conclusion

This technical document has provided a complete walkthrough of:

1. **Data Loading & Validation** - Successfully loaded 7 data files
2. **Data Transformation** - Implemented geospatial mapping and cost imputation
3. **Profit Calculation** - Merged sales and cost data with 95%+ success rate
4. **Visualization Generation** - Created 4 publication-ready figures
5. **Statistical Analysis** - Generated comprehensive metrics